Projection-Based Representative Learning for Image Classification in Heterogeneous Distributed Systems

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Unpublished working manuscript

Abstract

Distributed learning becomes particularly challenging when nodes differ simultaneously in their data distributions, model architectures, and learned representation spaces. We propose *Projection-Based Representatives (PBR)*, a Representative Learning approach for communicating image-format pseudo-data across heterogeneous nodes. PBR uses local Tucker decomposition to capture image structure, fine clustering to accommodate within-class heterogeneity, and predictive matching to adapt representative weights to the current local model while retaining the representative in the original image domain. We further incorporate finite historical memory and two-stream training to support repeated decentralized communication. Theoretical analysis relates representative fidelity to predictive discrepancy and within-cluster heterogeneity. Experiments on the DIGITS benchmark under simultaneous label, feature, and model heterogeneity show substantial improvements in cross-node generalization over prototype- and synthetic-anchor-based competitors. Ablation results further reveal a trade-off between architecture-specific latent matching and the interpretability and potential transferability of data-format representatives.

Keywords: Representative learning; distributed learning; data heterogeneity; model heterogeneity; image classification; pseudo-data.

Manuscript status: Unpublished working manuscript.

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1 Introduction

Distributed machine learning (DML) enables multiple nodes to learn collaboratively while keeping their raw observations local. Such systems must often operate under substantial heterogeneity in local data distributions, model architectures, computational resources, and communication schedules. Classical federated learning (FL) addresses data locality by training models locally and communicating model parameters or updates. However, communication through model-derived quantities becomes increasingly restrictive when local models or representation spaces are incompatible.

Representative Learning (RepL) provides a complementary, data-centric perspective on distributed learning (Li and Yang, 2022; Li, 2026). Rather than repeatedly communicating model parameters or gradients, each node constructs compact pseudo-data, called *representatives*, that preserve selected statistical or model-relevant information from its local observations while retaining the original data format. Representatives can therefore serve as reusable communication objects for downstream learning without requiring recipient nodes to share the same model architecture. This data-format communication and reuse are central motivations of the existing RepLframework.

Existing RepLconstructions have largely focused on structured data, progressing from moment-based and score-matching representatives to response-aided, transformed, anchored, and more recently distributional constructions (Li and Yang, 2022; Zheng et al., 2024; Li, 2026); extending this principle to images introduces a new challenge because a representative image should preserve meaningful structure in the original image domain while retaining information relevant to the current prediction task. Direct pixel-level averaging may fail to preserve important image structure, whereas communicating latent neural features ties the communication object to a particular encoder or representation space. The latter issue is especially important in decentralized learning when nodes employ heterogeneous model architectures.

Our contributions are threefold. First, we develop Projection-Based Representative for image data by combining Tucker-induced structure, model-aware predictive matching, and geometric anchoring in an image-format representative. Second, we integrate PBR with temporal-decay representative memory and local-representative mixture datasets training for decentralized learning under model and communication heterogeneity. Third, we establish local risk-matching and global alignment properties, as well as empirically study the roles of the individual PBR components through comparison and ablation experiments.

Figure 1 illustrates the central tension motivating our construction. PBR produces data-format representatives that retain recognizable image structure, whereas stronger matching in an architecture-specific latent space can yield less readily interpretable pseudo-images despite strong predictive performance within the participating network. This motivates our emphasis on balancing predictive fidelity with data-space structure and cross-model portability.

Our contributions are threefold. First, we develop PBR for image data by combining Tucker-induced structure, model-aware predictive matching, and geometric anchoring in a data-format representative. Second, we integrate PBR with finite representative memory and weighted two-stream training for decentralized learning under model and communication heterogeneity. Third, we establish local risk-approximation properties and empirically study the roles of the individual PBR components through comparison and ablation experiments.

The remainder of the paper is organized as follows. Section 2 reviews RepLand related



(a) Full PBR



(b) Ablation IV: latent-feature matching

Figure 1: Illustration of image-format representatives generated by the full PBR model and by architecture-specific latent-feature matching (Ablation IV) under the 10-outer-round × 100-local-epoch communication protocol. Each row corresponds to one node, and columns are organized by class and fine cluster; blank cells indicate unavailable class–cluster representatives. The comparison illustrates the distinction between preserving recognizable structure in the input domain and optimizing agreement in a model-dependent latent representation.

distributed-learning methods. Section 3 describes the distributed learning settings that motivate data-format representative communication. Section 4 develops PBR and its theoretical properties. Section 5 presents the simulation study and ablation analysis. Section 6 discusses the implications and limitations of the results, and the final section concludes the paper.

2 Background and Related Work

2.1 Representative Learning

Consider data distributed across K nodes,

$$\mathcal{D} = \bigcup_{k=1}^{K} \mathcal{D}_k, \quad \mathcal{D}_k = \{(\mathbf{x}_i, y_i) : i \in I_k\}, \tag{1}$$

where $n_k = |I_k|$. Representative Learning replaces repeated communication of model-derived quantities with compact weighted pseudo-data. More generally, node k may construct

$$\widetilde{\mathcal{D}}_k = \{(n_{kh}, \widetilde{y}_{kh}, \widetilde{\mathbf{x}}_{kh})\}_{h=1}^{r_k}, \tag{2}$$

where n_{kh} records the statistical weight associated with representative h. The representatives are constructed locally to preserve selected statistical, geometric, or model-related information from $\mathcal{D}_k$ while remaining in the original data format (Li and Yang, 2022; Li, 2026).

The earliest RepLconstructions are point-valued representatives. The *Mean Representative* (MR) provides a model-free summary through local covariate and response means (Li and Yang, 2022). The *Transformed Mean Representative* (TMR) modifies mean-based construction through link-function transformations to improve approximation under nonlinear generalized linear models (Li, 2026). In contrast, the *Score-Matching Representative* (SMR) explicitly preserves local score contributions, thereby incorporating local model information into the construction of the pseudo-observation (Li and Yang, 2022).

Subsequent constructions address limitations of score matching under heterogeneous or weakly identified local blocks. The *Response-Aided Score-Matching Representative* (RASMR) uses response- and model-related information to refine local partitions (Zheng et al., 2024), while *Anchored-SMR* augments score matching with an additional anchoring condition that resolves the underdetermined score system and preserves local geometry (Li, 2026). These methods show that RepLneed not be entirely model-free: a local model may guide representative construction while the communicated object remains a data-format pseudo-observation.

More recent developments extend RepLbeyond isolated point representatives. Lifted Empirical Risk Minimization replaces point observations with local probability distributions and evaluates expected loss over these distributions (Li and Wang, 2026). Clouded representatives use this perspective to retain local variability and geometric information that a single point cannot encode. Thus, the development of RepLproceeds from simple first-order summaries toward representatives preserving increasingly rich statistical and model-relevant structure.

2.2 Related Federated and Prototype-Based Learning

Most conventional federated-learning methods communicate model parameters or updates and therefore assume at least partial compatibility among local models. A different family of methods reduces communication by exchanging class-level prototypes or other compact feature summaries. FedProto, for example, communicates class prototypes in a learned representation space and uses prototype information to facilitate collaboration under heterogeneous local data (Tan et al., 2022).

A growing prototype-learning literature develops related mechanisms for handling data and system heterogeneity. FedSub (Campana and Delmastro, 2026) uses category-aware prototype information to identify client similarity and guide personalized subnetwork fusion; FedPLVM (Wang et al., 2024) incorporates local and global prototype clustering to account for cross-domain variability; DFPL (Zhang et al., 2025) considers decentralized prototype exchange; FedGMKD (Zhang et al., 2024) combines generated prototype features with knowledge distillation; and PPFPL (Zhang et al., 2025) studies privacy-preserving prototype communication. Other methods dynamically update prototypes during training to adapt the communicated feature summaries to heterogeneous local information.

Synthetic-data approaches move the communication object closer to the original input domain. DeSA (Huang et al., 2024) is particularly relevant to our setting because it introduces synthetic anchors for decentralized federated learning under both data and model heterogeneity. This makes DeSA conceptually closer to PBR than feature-prototype methods, as both communicate synthetic data-like objects that can be evaluated by heterogeneous local models.

PBR differs from both prototype communication and synthetic-anchor construction in the information it seeks to preserve. The representative is constrained through a data-derived Tucker representation and subsequently adapted to the predictive behavior of the current local model. Thus, model information guides representative construction without becoming the communication format itself.

Accordingly, we use FedProto (Tan et al., 2022) and DeSA (Huang et al., 2024) as the principal comparison methods in Section 5. FedProto provides a feature-space prototype baseline, whereas DeSA provides a synthetic data-space baseline. Among the candidate methods examined in our preliminary experiments, these two methods provided the strongest competing performance and were therefore selected for the main comparison.

2.3 From Structured to Image Data

The existing RepLconstructions motivate a general principle: the representative should preserve information important to downstream learning while remaining sufficiently independent of the recipient model to support distributed reuse. Applying this principle to images, however, introduces a new tension between statistical compression, semantic structure, and model compatibility.

A direct input-space construction takes the form

$$\widetilde{\mathbf{x}} = \sum_i w_i \mathbf{x}_i, \quad w_i \geq 0, \quad \sum_i w_i = 1. \tag{3}$$

Such a representative is independent of the learning architecture, but pixel-level averaging alone does not explicitly account for structured image variation.

Alternatively, suppose node k has an encoder ϕ_k. One may construct and communicate a latent

representative

$$\widetilde{\boldsymbol{\phi}}_k = \sum_i w_i \phi_k(\mathbf{x}_i). \tag{4}$$

This can provide an efficient task-relevant summary, but its meaning is tied to ϕ_k. When ϕ_k and ϕ_j differ, no common latent-coordinate interpretation is guaranteed.

PBR seeks an intermediate solution: construct the representative in a structured representation derived from the local data, use the current local model to determine predictive relevance, and communicate the final representative in the original image domain. Section 4 implements this principle using Tucker representations and predictive matching.

3 Distributed Learning Setting

The appropriate communication object depends on both the network topology and the compatibility of model representations across nodes. We distinguish three settings to clarify when latent representation communication is sufficient and when data-format representatives become necessary.

Centralized learning with a shared representation. Consider first a centralized network with a coordinating server and a common encoder ϕ available to all participating nodes. Node k may summarize its local observations directly in the shared representation space, for example,

$$\overline{\boldsymbol{\phi}}_k = \frac{1}{n_k} \sum_{i \in I_k} \phi(\mathbf{x}_i). \tag{5}$$

Because all participants interpret this vector through the same representation map, the representation itself can serve as the communication object. Reconstructing an input-space pseudo-image is unnecessary when the downstream computation operates entirely within the common feature space.

Decentralized learning with a shared representation. Now suppose that no central coordinator is available, but all nodes still share a common representation map ϕ. Nodes may communicate directly with their neighbors while exchanging representation-level summaries. Although the network topology changes, compatibility of the communicated objects is retained because the same encoder defines their meaning at every node. This setting can therefore be viewed as representation-level representative learning.

Decentralized learning with heterogeneous representations. The situation changes when node k employs model f_k with representation map ϕ_k, while node j employs f_j with

$$\phi_k \neq \phi_j. \tag{6}$$

A latent representation produced by node k is then defined relative to the internal representation of f_k and need not be directly interpretable by f_j. This incompatibility can arise from different architectures or from different locally learned representations.

In this setting, we communicate representatives in the original data domain. Node k constructs weighted pseudo-data

$$\mathcal{R}_{kh} = (n_{kh}, \widetilde{y}_{kh}, \widetilde{\mathbf{x}}_{kh}) \tag{7}$$

and exchanges these objects with connected nodes. Because $\widetilde{\mathbf{x}}_{kh}$ has the same input format as the original observations, each receiving node can process the representative through its own local model without requiring a shared latent coordinate system.

These settings can be viewed as a hierarchy of communication requirements. When a common representation space is available, communication may occur directly in that space. As model and representation heterogeneity increase, however, representation-level communication becomes less portable across nodes. Data-format representatives provide an alternative by moving the communication interface from the internal model representation back to the observable data domain.

The remainder of this paper focuses on the third setting: decentralized learning with heterogeneous local models and no assumed common encoder. Under this setting, the central methodological question is how to construct an image-format pseudo-observation that preserves local image structure while also carrying information relevant to the current local prediction task. Section 4 introduces PBR for this purpose.

4 Projection-Based Representatives (PBR)

RepLconstructs compact pseudo-data that preserve important information from local datasets while retaining the original data format. Existing representative constructions are primarily developed for structured data, where local information can be preserved through moments, transformations, or score equations. Extending this principle to image data is more challenging because direct pixel-level aggregation may fail to preserve meaningful image structure.

We therefore propose *Projection-Based Representatives (PBR)*. The main idea is to construct representatives in a structured representation space, while ensuring that the communicated object remains an image in the original data domain. In particular, we use Tucker decomposition to obtain a model-independent representation of image structure and subsequently refine the representative according to the predictive behavior of the current local model. Thus, the representation is determined by the data structure, whereas the representative weights are adapted to the current prediction task.

4.1 Local Tensor Representation and Fine Clustering

Consider node k with local image dataset

$$\mathcal{D}_k = \{(\mathbf{x}_i, y_i) : i \in I_k\}.$$

For each response class c, let

$$I_{k,c} = \{i \in I_k : y_i = c\}.$$

We first perform a Tucker decomposition locally within each class. For an image $\mathbf{x}_i$ belonging to

class c, write

$$\mathbf{x}_i \approx \mathcal{G}_i \times_1 U_{k,c,1} \times_2 U_{k,c,2}, \tag{8}$$

where $\mathcal{G}_i$ denotes the Tucker core and $U_{k,c,1}$ and $U_{k,c,2}$ are the corresponding factor matrices.

The Tucker decomposition is performed entirely locally. Neither the factor matrices nor individual Tucker cores are transmitted to other nodes. Instead, they provide a model-independent coordinate system for constructing local representatives.

Because observations within the same class may still exhibit substantial feature heterogeneity, the Tucker cores are further partitioned into fine clusters. Let q index such a cluster and suppose cluster q contains n_q observations sharing common label c. The resulting representative is constructed separately within each (c, q) block.

4.2 Tucker-Induced Representative Construction

For a fine cluster q containing n_q observations, define the simplex

$$\Delta = \left\{ w : w_i \geq 0,\ \sum_{i=1}^{n_q} w_i = 1 \right\}. \tag{9}$$

Let

$$\mathcal{G}_1, \ldots, \mathcal{G}_{n_q}$$

denote the Tucker cores associated with the observations in cluster q. For a weight vector $w \in \Delta$, define the representative Tucker core by

$$\tilde{\mathcal{G}}_q(w) = \sum_{i=1}^{n_q} w_i \mathcal{G}_i. \tag{10}$$

The corresponding representative image is reconstructed using the local Tucker factors,

$$\tilde{\mathbf{x}}_q(w) = \tilde{\mathcal{G}}_q(w) \times_1 U_{q,1} \times_2 U_{q,2}. \tag{11}$$

Since both the Tucker-core aggregation and reconstruction are linear under fixed factor matrices, the reconstructed representative can be viewed as a weighted combination of the locally reconstructed images. The simplex constraint therefore maintains an explicit connection between the representative and the local image cluster.

Define the unnormalized Tucker-core Gram matrix

$$[A_q]_{i\ell} = \langle \mathcal{G}_i, \mathcal{G}_\ell \rangle_F. \tag{12}$$

The matrix A_q characterizes the local geometry of the Tucker cores and will be used to regularize representative construction.

4.3 Predictive Matching

A representative should preserve not only the geometric structure of the local image cluster, but also its behavior under the current local model. At communication round r, let

$$p_i^{(r-1)} = f_{\theta_q^{(r-1)}}(\mathbf{x}_i) \tag{13}$$

denote the prediction vector for observation i under the previous local model.

Rather than forcing the representative to produce a perfect one-hot prediction, we seek to preserve the predictive behavior of the observations that it represents. For a given w, the target predictive response is therefore

$$\tilde{p}_q^{(r-1)}(w) = \sum_{i=1}^{n_q} w_i p_i^{(r-1)}. \tag{14}$$

We define the predictive discrepancy by

$$D_q^{(r)}(w) = \left\| f_{\theta_q^{(r-1)}}\left(\tilde{\mathbf{x}}_q(w)\right) - \sum_{i=1}^{n_q} w_i f_{\theta_q^{(r-1)}}(\mathbf{x}_i) \right\|_2^2. \tag{15}$$

This criterion differs from forcing $f_{\theta_q}(\tilde{\mathbf{x}}_q)$ toward the one-hot class response. Instead, the representative is required to reproduce the actual predictive behavior of the current local model over the represented observations.

4.4 PBR Optimization

Let

$$w_q^{(0)} = \left(\frac{1}{n_q}, \ldots, \frac{1}{n_q}\right)^\top \tag{16}$$

denote the equal-weight Tucker representative. The proposed PBR weights at communication round r are obtained from

$$w_q^{(r)} = \arg\min_{w \in \Delta} \left\{ D_q^{(r)}(w) + \gamma_q(w) \left(w - w_q^{(0)}\right)^\top A_q \left(w - w_q^{(0)}\right) \right\}. \tag{17}$$

The first term preserves the predictive behavior of the local cluster. The second term provides a global anchor toward the Tucker-core mean representative, with deviations measured according to the local Tucker geometry. This prevents predictive matching from producing a representative that is supported by only a small and potentially unrepresentative subset of the local observations.

We define the effective sample size of the representative by

$$\mathrm{ESS}_q(w) = \frac{1}{\sum_{i=1}^{n_q} w_i^2}. \tag{18}$$

Thus,

$$1 \le \mathrm{ESS}_q(w) \le n_q,$$

with $\mathrm{ESS}_q(w) = n_q$ under uniform weighting and decreasing values as the representative becomes concentrated on fewer observations.

To account for both Tucker-core scale and representative concentration, we set

$$\gamma_q(w) = \frac{1}{\lambda_{\max}(A_q)\,\mathrm{ESS}_q(w)}. \tag{19}$$

The resulting image

$$\tilde{\mathbf{x}}_q^{(r)} = \tilde{\mathbf{x}}_q\left(w_q^{(r)}\right) \tag{20}$$

is communicated together with its representative response and effective weight.

Design components. The proposed PBR construction contains three main structural components: (i) Tucker-induced reconstruction, which retains the representative in a data-derived image space; (ii) predictive matching, which incorporates information from the current local model without relying on an architecture-specific latent representation; and (iii) Tucker-space anchoring, which controls departure from the equal-weight representative. The contribution of these components is examined separately through ablation experiments in Section 5.

4.5 Representative Responses and Weights

Since all observations in a fine cluster share class label c, the representative may retain the hard class response

$$\tilde{y}_q = e_c. \tag{21}$$

Alternatively, when predictive uncertainty is to be retained, we may use the model-based response

$$\tilde{y}_q^{(r)} = f_{\theta_q^{(r-1)}}\left(\tilde{\mathbf{x}}_q^{(r)}\right). \tag{22}$$

The representative is therefore communicated as a weighted pseudo-observation

$$\mathcal{R}_q^{(r)} = \left(n_q^{(r)}, \tilde{y}_q^{(r)}, \tilde{\mathbf{x}}_q^{(r)}\right), \tag{23}$$

where $n_q^{(r)}$ denotes its effective statistical weight. The weight is determined by the amount of local information summarized by the representative and may account for the concentration of the optimized weights through $\mathrm{ESS}_q(w_q^{(r)})$.

For a newly constructed representative, we initialize its effective statistical weight as

$$n_q^{(r)} = n_q r(0) = n_q(1 - e^{-1}). \tag{24}$$

Unlike an ordinary sample count, $n_q^{(r)}$ need not be an integer. It is interpreted as the statistical mass carried by the pseudo-observation and enters downstream learning through a weighted loss. This interpretation allows representatives generated at different communication rounds to be retained with different effective weights.

4.6 Representative Memory and Mixture Training

Temporal-decay representative memory. At communication round r, a host node k may have received multiple representatives from the same guest node at different communication rounds. Rather than replacing an older representative whenever a new one arrives, we retain a finite memory of historical representatives. Let

$$\mathcal{H}_k^{(r)} = \{\mathcal{R}_h = (n_h, \tilde{y}_h, \tilde{\mathbf{x}}_h)\}_{h=1}^{H_k^{(r)}} \tag{25}$$

denote the representative memory available at node k at round r.

Retaining historical representatives is useful because repeated representative construction need not yield the same pseudo-observation. The predictive-matching objective in (17) is generally nonconvex, and different communication rounds or numerical initializations may lead to different representative solutions. Historical representatives therefore provide a finite-memory collection of previously communicated local summaries rather than requiring each new representative to completely replace its predecessors.

Let $\tau \geq 0$ denote the age of a representative. We use geometric decay

$$r(\tau) = (1 - e^{-1})e^{-\tau}, \quad \tau = 0, 1, \ldots. \tag{26}$$

The effective weight of an age-τ representative originating from a block with weight n_q is therefore

$$n_q^{(\tau)} = n_q r(\tau). \tag{27}$$

Since

$$\sum_{\tau=0}^{\infty} r(\tau) = 1, \tag{28}$$

the decay is normalized so that, in the absence of truncation,

$$\sum_{\tau=0}^{\infty} n_q r(\tau) = n_q. \tag{29}$$

Thus, the decay distributes a fixed nominal statistical mass across representative ages; truncation at ϵ removes negligible residual mass. Moreover,

$$r(0) = 1 - e^{-1} > (1 - e^{-1})e^{-1} = r(1), \tag{30}$$

so the newly received representative receives greater weight than any individual historical version.

A historical representative is retained while

$$n_q^{(\tau)} \geq \epsilon, \tag{31}$$

where $\epsilon > 0$ is a memory threshold. Representatives whose effective weights fall below ϵ are removed from the memory. The resulting mechanism therefore provides finite representative memory with decreasing influence as information becomes older.

Historical memory also allows representative diversity across communication rounds. While the regularization anchor in (17) remains the equal-weight vector $w_q^{(0)}$, the numerical optimization may be initialized from different points in Δ across rounds. This distinction is important: $w_q^{(0)}$ defines the fixed global anchor of the objective, whereas the numerical starting value determines the trajectory used to search the nonconvex objective. Retaining historical representatives therefore preserves potentially distinct local solutions generated across communication rounds.

Local-representative mixture training. At host node k, the original local observations and the representatives stored in $\mathcal{H}_k^{(r)}$ form two distinct sources of training information. Let

$$n_{-k} = \sum_{h \in \mathcal{H}_k} n_h \tag{32}$$

denote the total effective statistical mass of the current representative memory.

For a local training step, let B_L denote the local-data batch mass and B_R the representative batch mass. A minibatch $\mathcal{B}_k^L \subset \mathcal{D}_k$ containing B_L local observations is sampled from the original data. Because representative weights need not be integers, representatives are not replicated according to n_h. Instead, the representative batch mass B_R is distributed proportionally across the stored representatives:

$$b_h = B_R \frac{n_h}{\sum_{j \in \mathcal{H}_k^{(r)}} n_j}, \quad h \in \mathcal{H}_k^{(r)}. \tag{33}$$

Consequently,

$$\sum_{h \in \mathcal{H}_k^{(r)}} b_h = B_R, \tag{34}$$

and the individual b_h may be non-integer.

The local model is updated by minimizing

$$L_k^{(b)}(\theta_k) = \frac{1-\alpha_k}{B_L} \sum_{i \in \mathcal{B}_k^L} \ell\left(\theta_k; \mathbf{x}_i, y_i\right) + \frac{\alpha_k}{B_R} \sum_{h \in \mathcal{H}_k^{(r)}} b_h \ell\left(\theta_k; \tilde{\mathbf{x}}_h, \tilde{y}_h\right), \tag{35}$$

where $\alpha_k \in [0,1]$ controls the overall contribution of information received from other nodes relative to the local data.

The formulation separates three conceptually different quantities. The minibatch masses (B_L, B_R) control the computational allocation between the two training streams; the representative weights n_h describe the effective statistical information carried by individual pseudo- observations; and α_k controls the overall strength of cross-node collaboration. Thus, a representative with a large n_h need not be physically repeated many times in a minibatch. Its influence is represented through the continuous loss weight b_h.

Communication cycle. After local training, each node reconstructs its current representatives using the updated local model and broadcasts the resulting weighted pseudo-observations to its connected neighbors. Newly received representatives are added to the recipient's representative

memory, while existing historical representatives are aged according to (26). Local training, representative reconstruction, communication, and memory updating are repeated across communication rounds. Algorithm 1 summarizes the complete procedure.

4.7 Theoretical Properties of PBR

We next establish two basic properties of the proposed PBR construction. The first characterizes the geometry of the reconstructed representative and quantifies the stabilizing effect of Tucker-space anchoring. The second establishes predictive fidelity relative to the weighted local observations. These results concern representative construction at a fixed communication round and therefore do not require convexity of the underlying neural network training problem.

Proposition 1 (Geometric validity and anchored deviation)**.** *Suppose the Tucker factors $U_{q,1}$ and $U_{q,2}$ are fixed within fine cluster q, and define*

$$\widehat{\mathbf{x}}_i = \mathcal{G}_i \times_1 U_{q,1} \times_2 U_{q,2}, \quad i = 1, \dots, n_q. \tag{36}$$

Then, for every $w \in \Delta$,

$$\widetilde{\mathbf{x}}_q(w) = \sum_{i=1}^{n_q} w_i \widehat{\mathbf{x}}_i \in \operatorname{conv}\left\{\widehat{\mathbf{x}}_1, \dots, \widehat{\mathbf{x}}_{n_q}\right\}. \tag{37}$$

Moreover,

$$(w - \bar{w}_q)^\top A_q (w - \bar{w}_q) = \left\| \widetilde{\mathcal{G}}_q(w) - \bar{\mathcal{G}}_q \right\|_F^2. \tag{38}$$

If the Tucker factors have orthonormal columns, then

$$\left\| \widetilde{\mathbf{x}}_q(w) - \bar{\mathbf{x}}_q^T \right\|_F^2 = (w - \bar{w}_q)^\top A_q (w - \bar{w}_q). \tag{39}$$

The purpose of predictive matching is ultimately not only to reproduce model outputs, but also to preserve the local training objective. We therefore quantify the discrepancy between the loss evaluated at an PBR representative and the weighted empirical loss of the observations represented by it.

For a fixed communication round, suppress the round index and define

$$p_i = f_\theta(\mathbf{x}_i), \quad \bar{p}_q(w) = \sum_{i=1}^{n_q} w_i p_i, \quad \widetilde{p}_q(w) = f_\theta\left(\widetilde{\mathbf{x}}_q(w)\right), \tag{40}$$

so that

$$D_q(w) = \|\widetilde{p}_q(w) - \bar{p}_q(w)\|_2^2. \tag{41}$$

We further define the within-cluster predictive dispersion

$$V_q(w) = \sum_{i=1}^{n_q} w_i \|p_i - \bar{p}_q(w)\|_2^2. \tag{42}$$

Algorithm 1 PBR with two-stream training

Require: Local datasets $\{\mathcal{D}_k\}_{k=1}^K$; communication rounds T; local epochs E; $\alpha_k = 1/2$; memory threshold $\epsilon = 1$; number of classes $K_y = 10$.

Ensure: Trained local models $\{\theta_k^{(T)}\}_{k=1}^K$.

for all nodes k **do**
  Initialize θ_k on $\mathcal{D}_k$ and set $\mathcal{H}_k \leftarrow \varnothing$. ▷ Initial local models based on local data
  for all classes c **do**
    Compute the Tucker representation $x_{kci} \approx \mathcal{G}_{kci} \times_1 U_{k,1} \times_2 U_{k,2}$, and cluster the Tucker cores into clusters $q = 1, \ldots, Q_k$.
    for all cluster q **do**
      Compute $[A_{k,c,q}]_{i\ell} \leftarrow \langle \mathcal{G}_{kci}, \mathcal{G}_{kc\ell} \rangle_F$.
      Initial $w_{k,c,q} = w_{k,c,q}^{(0)}$.
      Construct $\tilde{x}_{k,c,q} = \left(\sum_{i=1}^{n_{k,c,q}} w_i \mathcal{G}_{kci}\right) \times_1 U_{k,1} \times_2 U_{k,2}$.
      Update $w_{k,c,q}$ by (17), and $\tilde{x}_{k,c,q}(w)$ accordingly.
      Construct $\mathcal{R}_{k,c,q} = \left(n_{k,c,q}(1 - e^{-1}), \tilde{y}_{k,c,q}, \tilde{x}_{k,c,q}\right)$, where $y_{k,c,q} = f_{\theta_k}(\tilde{x}_{k,c,q})$
    end for
  end for
  Broadcast representatives $\{\mathcal{R}_{k,c,q}\}_{c,q}$ to other connected nodes.
end for
for $t = 1, 2, \ldots, T$ **do** ▷ Iterative update
  for all nodes k **do** ▷ Update local model based on local data and representatives.
    Compute $n_{-k} = \sum_h n_h$.
    Initialize $m_k \leftarrow n_k$ and $m_h \leftarrow n_h$ for each $\mathcal{R}_h \in \mathcal{H}_k$, set $m_{-k} = n_{-k}$
    for $e = 1, 2, \ldots, E$ **do**
      Randomly generate B_L and B_R from hyper-geometry of (m_k, m_{-k})
      Randomly draw $\mathcal{B}_k^L = \{(x_i, y_i)\}_{i=1}^{B_L}$ from $\mathcal{D}_k$, update $m_k \leftarrow m_k - B_L$.
      Take $\mathcal{B}_k^R = \{(b_h, \tilde{y}_h, \tilde{x}_h)\}_{h=1}^H$ with $b_h = B_R n_h / \sum_{j=1}^H n_j$, update $m_h \leftarrow m_h - b_h$ and m_{-k}.
      Update θ_k by minimizing

$$L_k^{(b)} = \frac{1 - \alpha_k}{B_L} \sum_{i \in \mathcal{B}_k^L} \ell_i + \frac{\alpha_k}{B_R} \sum_{h \in \mathcal{B}_k^R} b_h \ell_h.$$

    end for
    for all class-clusters (c, q) **do** ▷ Construct representatives
      Initial $w_{k,c,q} = w_{k,c,q}^{(0)}$.
      Construct $\tilde{x}_{k,c,q} = \left(\sum_{i=1}^{n_{k,c,q}} w_i \mathcal{G}_{kci}\right) \times_1 U_{k,1} \times_2 U_{k,2}$.
      Update $w_{k,c,q}$ by (17), and $\tilde{x}_{k,c,q}(w)$ accordingly.
      Construct $\mathcal{R}_{k,c,q}$.
    end for
    for all $\mathcal{R}_h = (n_h, \tilde{y}_h, \tilde{x}_h) \in \mathcal{H}_k$ **do** ▷ Age representative
      Update $n_h \leftarrow n_h e^{-1}$.
      if $n_h < \epsilon$ **then**
        $\mathcal{H}_k \leftarrow \mathcal{H}_k \setminus \{\mathcal{R}_h\}$.
      end if
    end for
    Broadcast representatives $\{\mathcal{R}_{k,c,q}\}_{c,q}$ to other connected nodes.
  end for
  Evaluate $\{\theta_k^{(t)}\}_{k=1}^K$.
end for

Theorem 1 (Local risk preservation)**.** *Let $\ell(p, c)$ denote the classification loss for prediction vector p and common cluster label c. Suppose that $\ell(\cdot, c)$ is L_1-Lipschitz and has L_2-Lipschitz gradient with respect to its prediction argument. Then, for any $w \in \Delta$,*

$$\left| \ell\left(\widetilde{p}_q(w), c\right) - \sum_{i=1}^{n_q} w_i \ell(p_i, c) \right| \leq L_1 \sqrt{D_q(w)} + \frac{L_2}{2} V_q(w). \tag{43}$$

Theorem 1 identifies two sources of representative error. The first is the predictive discrepancy $D_q(w)$ directly minimized by PBR, while the second is the predictive heterogeneity $V_q(w)$ within the fine cluster. Hence, PBR provides an accurate local risk approximation when the representative successfully matches the weighted prediction and the fine clusters are locally homogeneous. This also provides a theoretical motivation for fine clustering before representative construction.

We next connect representative fidelity to the optimization error induced by replacing local observations with PBR pseudo-data. Let the full-data objective be

$$F(\theta) = \sum_{k=1}^{K} \frac{n_k}{N} F_k(\theta), \quad N = \sum_{k=1}^{K} n_k, \tag{44}$$

and let $\widetilde{F}_k(\theta)$ denote the corresponding local objective constructed from the PBR representatives at node k.

Theorem 2 (Convergence under PBR approximation)**.** *Suppose that F is μ-strongly convex and L-smooth. Assume that, for each node k, the PBR surrogate satisfies*

$$\left\| \nabla F_k(\theta) - \nabla \widetilde{F}_k(\theta) \right\|_2 \leq \varepsilon_k \tag{45}$$

for all θ in the optimization region, and define

$$\varepsilon = \sum_{k=1}^{K} \frac{n_k}{N} \varepsilon_k. \tag{46}$$

Consider the representative-based update

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \widetilde{F}\left(\theta^{(t)}\right), \tag{47}$$

where

$$\widetilde{F}(\theta) = \sum_{k=1}^{K} \frac{n_k}{N} \widetilde{F}_k(\theta). \tag{48}$$

For a sufficiently small step size η, the iterates satisfy

$$\left\| \theta^{(T)} - \theta^\star \right\|_2^2 \leq (1 - c\eta\mu)^T \left\| \theta^{(0)} - \theta^\star \right\|_2^2 + O\left(\frac{\eta}{\mu} \varepsilon^2\right), \tag{49}$$

for some constant $c > 0$, where $\theta^\star$ is the minimizer of F.

Theorem 2 shows that representative-based learning approaches the full-data solution up to an

error determined by the quality of the PBR approximation. Together with Theorem 1, the result suggests that finer and more homogeneous clusters, together with smaller predictive discrepancy, improve representative fidelity and therefore reduce the error introduced into distributed learning.

The strong-convexity assumption in Theorem 2 is used to characterize the effect of representative approximation independently of neural-network nonconvexity. The theorem should therefore be interpreted as a surrogate optimization result rather than a global convergence guarantee for the CNN and AlexNet models used in Section 5.

5 Simulation

We evaluated FedProto, DeSA, and PBR on DIGITS under a two-stage heterogeneous partition designed to induce both label and feature heterogeneity. All methods used identical data partitions, model assignments, initialization, and local-training budgets.

5.1 Experimental Setup

Dataset We evaluate the proposed representative-learning method on the DIGITS benchmark used by DESA. The benchmark contains five visually heterogeneous digit domains: MNIST, SVHN, USPS, SynthDigits, and MNIST-M.

The five DIGITS domains were pooled and repartitioned using a class-stratified 8:2 training-test split with seed 2026. All images were converted to three-channel 32×32 inputs and normalized using domain-specific statistics.

Heterogeneous data partition. The pooled DIGITS training set was allocated to $K = 10$ nodes through a two-stage heterogeneous partitioning procedure designed to induce both label and feature heterogeneity.

In the first stage, label heterogeneity was generated using a Dirichlet distribution with concentration parameter $\alpha_y = 0.1$. For each class c, let N_c denote the total number of observations from class c. We sampled $(\pi_{1c}, \ldots, \pi_{Kc}) \sim \text{Dirichlet}(\alpha_y, \ldots, \alpha_y)$, and determined the number of class-c observations allocated to node k, n_{kc}, according to these proportions, subject to $N_c = \sum_{k=1}^{K} n_{kc}$. This stage determines only the class counts assigned to each node.

In the second stage, feature heterogeneity was introduced through Tucker-core representations with spatial rank 8×8. For each class c, a Tucker decomposition was performed on all training observations belonging to that class,

$$x_i \approx \mathcal{G}_i \times_1 U_{1,c} \times_2 U_{2,c},$$

where $\mathcal{G}_i$ denotes the Tucker core tensor of image x_i.

The resulting Tucker cores were clustered into three latent groups using K-means ($K = 3$) in the Tucker representation space. A second Dirichlet distribution with concentration parameter $\alpha_\phi = 0.1$ was then used to determine the proportions of latent groups assigned to each node. Given the class sample sizes n_{kc} determined in Stage 1, observations were sampled from the latent Tucker groups accordingly. This procedure induces feature and representation heterogeneity while preserving the class counts determined by the first-stage label allocation.

Therefore, Stage 1 controls the node-specific label distributions $P_k(Y)$, whereas Stage 2 controls the class-conditional feature distributions $P_k(X|Y)$.

The class-specific Tucker bases and K-means centroids were fitted using the training data only. Test samples were projected using these training-derived representations and allocated to the nodes according to their class and class-conditional latent-group distributions. Thus, the test sets follow the corresponding training heterogeneity without being used to construct the partition.

Decentralized network. The communication graph was fully connected. At each communication round, every node constructed representatives from its original local training data and transmitted the resulting representatives to the other nine nodes. Raw local observations and model parameters were never transmitted.

Model architectures and optimization. Five nodes used ConvNet and the remaining five used AlexNet. ConvNet consisted of three convolutional blocks, each containing a 128-channel convolution, group normalization, ReLU activation, and average pooling, followed by a 10-class linear classifier. AlexNet contained five convolutional layers with 128, 192, 256, 192, and 192 output channels, respectively, followed by a 10-class linear classifier.

All models were implemented in PyTorch and optimized using SGD with a learning rate of 0.01, momentum 0.5, and a batch size of 64.

5.1.1 Communication Regimes

To investigate the impact of communication frequency, we considered three communication regimes while maintaining a total of 1,000 local training iterations.

i. **Frequent communication:** 1,000 outer iterations with one local iteration between communications;

ii. **Regular communication:** 100 outer iterations with 10 local iterations between communications;

iii. **Infrequent communication:** 10 outer iterations with 100 local iterations between communications.

These settings retain the same total local-training budget while varying the frequency of information exchange. Results were evaluated at communication rounds 0, 1, 10, 100, and 1,000 whenever the corresponding communication regime contained that round. Round 0 denotes the common local initialization shared by all methods and communication protocols.

5.2 Methods and Evaluation

5.2.1 Comparison Methods

FedProto used class-level penultimate-layer prototypes. Because the heterogeneous Dirichlet partition may leave some nodes without observations from certain classes, a node in DeSA generated synthetic anchors only for its locally observed classes. The global anchor for each class was obtained by averaging the corresponding anchors over nodes containing that class.

5.2.2 Representative Construction

For all RepL variants, a node-class block containing fewer than 20 observations did not participate in representative construction. Eligible samples were clustered into three groups, and clusters containing fewer than five observations were merged into their nearest feature-centroid cluster.

Representative weights were constrained to have an effective sample size of at least five. If the optimized weights yielded $\mathrm{ESS}_{kc} < 5$, the weights were minimally shrunk toward the uniform vector until $\mathrm{ESS}_{kc} \geq 5$.

A representative was transmitted only if its source model predicted the assigned class correctly. For correctly classified representatives, let $p_c(\tilde{\mathbf{x}}_{kc})$ denote the source model's predicted probability for the representative's true class. The confidence adjustment was

$$q_{kc} = \begin{cases} 1, & p_c(\tilde{\mathbf{x}}_{kc}) \geq 0.5, \\ \exp(-1), & p_c(\tilde{\mathbf{x}}_{kc}) < 0.5. \end{cases} \tag{50}$$

Thus, correctly classified but low-confidence representatives remained available to receiving nodes but contributed less to the local classification loss.

5.2.3 Learning with Representatives

PBR was implemented using the historical-memory and two-stream training procedure described in Section 4 and Algorithm 1. Each node retained both newly received and historical representatives. Historical representative weights were exponentially decayed by a factor of e^{-1} after each communication round, and representatives with effective weight below $\epsilon = 1$ were removed. Local and representative information were combined using $\alpha_k = 1/2$. Representative contributions within each minibatch were allocated proportionally to their current effective weights.

5.2.4 Ablation Models

To isolate the contribution of the main components of PBR, we consider four ablation models.

Ablation I: No global anchor. We first remove the Tucker-space anchoring penalty while retaining the same convex Tucker representative:

$$w_q^{(r)} = \arg\min_{w \in \Delta} \left\| f_{\theta_q^{(r-1)}}(\tilde{\mathbf{x}}_q(w)) - \sum_i^{n_q} w_i f_{\theta_q^{(r-1)}}(\mathbf{x}_i) \right\|_2^2. \tag{51}$$

This gives the role of anchoring elsewhere in RepL as a mechanism for preserving representative geometry and improving stability.

Ablation II: Image-space convex representative. To isolate the contribution of Tucker-space reconstruction, we replace the Tucker representative by a direct convex combination of the original images, $\tilde{\mathbf{x}}_q = \sum_{i=1}^{n_q} w_i \mathbf{x}_i$. The representative weights are then obtained from

$$w_q^{(r)} = \arg\min_{w \in \Delta} \left\{ \left\| f_{\theta_q^{(r-1)}}(\tilde{\mathbf{x}}_q(w)) - \sum_i^{n_q} w_i f_{\theta_q^{(r-1)}}(\mathbf{x}_i) \right\|_2^2 + \gamma \left\| w - w_q^{(0)} \right\|_2^2 \right\}, \tag{52}$$

with $\gamma = 1/\operatorname{ESS}_q(w)$.

Ablation III: Unconstrained synthetic representative. To evaluate the importance of the convex-combination constraint, we remove the weight parameterization entirely and optimize the representative image directly. Let

$$\tilde{\mathbf{x}}_q^{(r)} = \arg\min_{\mathbf{x}} \left\| f_{\theta_q^{(r-1)}}(\mathbf{x}) - \frac{1}{n_q} \sum_{i=1}^{n_q} p_i^{(r-1)} \right\|_2^2. \tag{53}$$

Ablation IV: Latent-feature matching. Finally, we replace architecture-independent prediction matching with matching in the local model's latent feature space. Let $h_{\theta_q^{(r-1)}}(\mathbf{x})$ denote the activation vector from the layer immediately preceding the final classification layer. We replace the predictive discrepancy in the full model by

$$w_q^{(r)} = \arg\min_{w \in \Delta} \Bigg\{ \left\| h_{\theta_q^{(r-1)}}(\tilde{\mathbf{x}}_q(w)) - \sum_i^{n_q} w_i h_{\theta_q^{(r-1)}}(\mathbf{x}_i) \right\|_2^2 + \gamma \left(w - w_q^{(0)} \right)^\top A_q \left(w - w_q^{(0)} \right) \Bigg\}, \tag{54}$$

where $\gamma = \frac{s_q}{\lambda_{\max}(A_q) \operatorname{ESS}_q(\mathbf{w})}$, $s_q = \frac{1}{n_q} \sum_{i=1}^{n_q} \left\| h_{\theta_q^{(r-1)}}(\mathbf{x}_i) \right\|_2^2$. This final ablation evaluates whether architecture-specific latent features provide additional information beyond predictive-output matching.

5.2.5 Evaluation Protocol

We evaluate each local model using both *own-node* and *cross-node* test accuracy. Own-node accuracy measures the degree to which collaboration preserves performance on the node's local task, whereas cross-node accuracy evaluates the ability of the local model to generalize across heterogeneous node distributions.

For node k, the own-node accuracy is

$$\operatorname{Acc}_k^{\text{own}} = \operatorname{Acc}\left(M_k, D_k^{\text{test}}\right), \tag{55}$$

which measures predictive performance on the test distribution corresponding to the node's own local training distribution.

To evaluate generalization beyond the local data distribution, we additionally compute cross-node accuracy. For model k, define

$$\operatorname{Acc}_k^{\text{cross}} = \frac{1}{\sum_{j \neq k} n_j} \sum_{j \neq k} n_j \operatorname{Acc}\left(M_k, D_j^{\text{test}}\right). \tag{56}$$

Thus each node-level cross-node accuracy is the weighted mean accuracy of model M_k across all other nine node-specific test sets.

Round 0 denotes the common local initialization shared by all methods and communication protocols. In the result tables, C and A denote ConvNet and AlexNet, respectively. Boldface

indicates the proposed PBR result among the compared methods.

5.3 Results

We first examine cross-node generalization performance. Table 1 reports the weighted mean cross-node test accuracy of FedProto, DeSA, the proposed PBR method, and the four PBR ablation variants under the three communication regimes. Detailed node-specific cross-node and own-node results are provided in Tables S2–S3 in the Supplementary Materials.

Table 1: Weighted mean cross-node generalization accuracy (%) on DIGITS under different communication regimes.

Protocol	Round	FedProto	DeSA	PBR	Abl. I	Abl. II	Abl. III	Abl. IV
Shared initialization	0				30.87			
100 outer × 10 local	1	33.31	36.99	**38.81**	38.90	38.75	38.32	37.64
	10	35.49	37.18	**52.38**	52.08	49.39	44.25	56.19
	100	36.14	33.86	**59.25**	59.60	52.86	46.06	77.91
1000 outer × 1 local	1	31.10	34.27	**32.01**	31.95	32.13	32.37	30.10
	10	33.57	36.76	**41.18**	41.33	41.11	37.63	36.38
	100	35.50	37.44	**52.71**	52.41	49.84	43.82	59.27
	1000	36.10	34.53	**63.37**	64.20	54.48	46.39	81.43
10 outer × 100 local	1	35.53	37.12	**45.99**	46.08	45.69	44.80	46.74
	10	36.13	33.38	**54.43**	53.29	50.03	45.94	70.23

Note: For each node k, cross-node accuracy is the weighted mean accuracy of model M_k over the other nine node-specific test sets, with weights proportional to the corresponding target test-set sizes. The reported weighted mean further aggregates these cross-node accuracies across the ten host models. Boldface identifies the proposed PBR model rather than the largest entry within each row.

Overall, PBR substantially improves cross-node generalization relative to FedProto and DeSA once representative exchange has progressed. For example, under the 100 outer × 10 local setting, PBR increases from 38.81% at Round 1 to 52.38% at Round 10 and 59.25% at Round 100, whereas FedProto and DeSA remain below 40% at Round 100. A similar pattern is observed under frequent communication (1000 outer × 1 local), where PBR reaches 63.37% at Round 1000, compared with 36.10% for FedProto and 34.53% for DeSA. Under the less frequent 10 outer × 100 local regime, PBR achieves 54.43% at the final communication round, again exceeding both comparison methods.

The gain in cross-node generalization is accompanied by a modest reduction in aggregate own-node accuracy relative to FedProto and DeSA, indicating a trade-off between local specialization and generalization across heterogeneous node distributions. The detailed results in Table S3 further show that this trade-off is heterogeneous across nodes: some local models retain high own-node accuracy while others sacrifice more local performance as cross-node generalization improves.

The ablation results clarify the roles of the individual components of PBR. Removing the global Tucker-space anchor (Ablation I) generally produces performance close to the full model, indicating that the predictive-matching criterion accounts for a substantial portion of the improvement, while the anchor primarily provides additional geometric control and stabilization. Replacing Tucker-induced reconstruction with a direct image-space convex representative (Ablation II) results in larger performance losses at later rounds. Removing the convex-combination constraint entirely (Ablation III) produces the weakest PBR variant in most later-round settings. Together, these

results support retaining an explicit connection between the representative and the local image geometry.

Ablation IV exhibits a notably different pattern. It replaces predictive output matching with matching of the activation vector immediately preceding the classification layer. As shown in Table 1, Ablation IV substantially outperforms the full PBR model after sufficient communication. Its weighted cross-node accuracy reaches 77.91%, 81.43%, and 70.23% under the 100×10, 1000×1, and 10×100 communication protocols, respectively, compared with 59.25%, 63.37%, and 54.43% for the full PBR model. Thus, architecture-specific latent-feature matching can be highly effective among the nodes participating in distributed training.

We conjecture, however, that this gain may partly reflect a new form of overfitting, which we refer to as *distributed structure overfitting*. Unlike conventional overfitting to individual training observations, distributed structure overfitting occurs when communicated pseudo-data become specialized to the collection of models and representations participating in the distributed system. Repeated communication may then allow the participating models and communicated representatives to become increasingly compatible with one another, yielding high cross-node accuracy within the existing network without necessarily preserving transferability to a previously unseen node.

This concern is particularly relevant to Ablation IV because its matching criterion is defined through $h_{\theta_q}(\mathbf{x})$, an architecture- and model-dependent latent representation. A representative can therefore be highly informative to the models whose latent spaces contributed to its construction while being less meaningful to a newly introduced model with a different architecture, initialization, or learned representation. In contrast, the full PBR model performs matching in the prediction space while retaining Tucker-induced structure in the original image domain, thereby maintaining a stronger connection between the communicated representative and the underlying data.

The representative images provide qualitative evidence consistent with this interpretation. Figure 1 compares the complete sets of representatives produced by the full PBR model and Ablation IV at the final round under the infrequent-communication protocol (10 outer rounds $\times$ 100 local epochs). Each row in a panel corresponds to one node, and the columns are organized by class and fine cluster. Blank cells indicate class–cluster representatives that are unavailable at the corresponding node.

Although Ablation IV achieves substantially higher cross-node accuracy among the participating nodes, the visual comparison reveals an important trade-off. Representatives produced by the full PBR model generally retain recognizable digit structure in the original image space, whereas some representatives produced by Ablation IV are visually ambiguous with respect to their assigned classes. Thus, strong performance within the participating network does not necessarily imply equally interpretable or model-independent communicated pseudo-data. We discuss this phenomenon further in Section 6.

For completeness, representative images produced by all methods under all three communication protocols are reported in Figures S1, S2, and S3 of the Supplementary Materials.

Finally, comparison across communication regimes indicates that PBR benefits from repeated representative exchange. The improvement with increasing communication rounds is considerably larger for PBR and the latent-feature variant than for FedProto and DeSA. This behavior is consistent with the role of reusable representative pseudo-data in RepL: successive communication rounds allow local models to incorporate information from heterogeneous nodes without directly exchanging model parameters.

6 Discussion and Conclusion

6.1 Distributed Structure Overfitting

One of the most striking findings is that Ablation IV, which replaces prediction matching with architecture-specific latent-feature matching, substantially outperforms the full PBR model in cross-node prediction after sufficient communication. At the same time, Figure **??** shows that some of the resulting representatives become less visually distinguishable with respect to their assigned classes in the original image space.

We conjecture that this behavior may reflect a new form of overfitting, which we call *distributed structure overfitting*. Unlike conventional overfitting, which concerns specialization to a finite training sample, distributed structure overfitting concerns specialization of communicated information to the collection of models and representation spaces participating in the distributed system. Repeated communication may allow the participating models and communicated representatives to become increasingly compatible with one another, producing strong cross-node performance within the existing network without necessarily preserving transferability beyond that network.

Ablation IV is particularly susceptible to this mechanism because its matching objective is defined in the model-dependent space $h_{\theta_k}(\mathbf{x})$. A pseudo-image can therefore carry information that is highly effective for the participating models even if its semantic content is less apparent in the original image space. In contrast, the full PBR construction matches predictive behavior while retaining Tucker-induced structure, maintaining a more explicit connection between the communicated image and the source data geometry.

The present experiment provides evidence consistent with this hypothesis, but does not establish distributed structure overfitting because all models used for cross-node evaluation participated in the distributed learning process. A direct test would hold out an *unseen node* from representative construction and communication, then introduce the node after training and measure its ability to learn from the fixed representatives. Under the distributed-structure-overfitting hypothesis, the advantage of Ablation IV over the full PBR model should decrease when representative transfer is evaluated outside the distributed structure in which the pseudo-data were generated.

6.2 Representative Memory and Concentration

Historical representatives serve a role beyond retaining stale information. Because PBR construction involves a generally nonconvex predictive-matching objective, representatives generated at different communication rounds may correspond to different feasible solutions and may therefore contain complementary information. Finite historical memory allows these solutions to coexist temporarily, while temporal decay gradually reduces the influence of older representatives. This provides a natural balance between representative diversity and adaptation to the current local models.

The effective sample size (ESS) provides a second mechanism for controlling how local information is summarized. In this work, we use

$$\mathrm{ESS}_2(w) = \frac{1}{\sum_i w_i^2}, \tag{57}$$

which decreases as representative construction becomes concentrated on a small number of local

observations. This choice is closely related to a more general family based on Rényi entropy,

$$H_\nu(w) = \frac{1}{1-\nu} \log \left(\sum_i w_i^\nu \right), \tag{58}$$

with effective diversity

$$N_\nu(w) = \exp\{H_\nu(w)\}. \tag{59}$$

The current ESS corresponds to $N_2(w)$, whereas the limit $\nu \to 1$ gives the Shannon-entropy effective number

$$N_1(w) = \exp \left\{ -\sum_i w_i \log w_i \right\}. \tag{60}$$

Different choices therefore provide different notions of representative concentration. Investigating how these alternatives affect representative stability, statistical weighting, and downstream learning remains an open direction.

6.3 Limitations and Future Directions

Several limitations suggest natural extensions of PBR. First, Tucker decomposition is one choice of a structured, model-independent image representation rather than an essential restriction of the PBR principle. Alternative representations, including matrix or tensor factorizations, dictionary-based representations, and unsupervised latent constructions, could replace the Tucker cores provided that representative reconstruction back to the original data domain remains available.

Learned encoder-decoder representations provide another possibility. Such models may produce more flexible nonlinear representations and reconstructions, but they introduce dependence on the architecture and training of the encoder-decoder system. This creates a trade-off between representation flexibility and the model independence that motivates data-format communication in heterogeneous distributed systems.

Second, the representative response provides another design choice. The present framework can communicate a hard class response or a model-derived response. More generally, the weighted source prediction

$$\sum_i w_i f_\theta(\mathbf{x}_i) \tag{61}$$

can retain additional predictive uncertainty. Determining when hard versus soft representative responses yield better transfer across heterogeneous models remains an open problem.

Third, the theoretical results characterize local risk preservation and representative approximation under regularity conditions, while the CNN and AlexNet experiments involve nonconvex neural-network optimization. A complete convergence theory for PBR with heterogeneous local architectures, historical memory, and two-stream stochastic training remains to be developed.

Finally, the current PBR construction remains point-valued. A natural extension is to replace a representative image by a structured distributional representative that also retains local uncertainty

and within-cluster variability. Such an extension would connect PBR with the broader development of distributional or clouded representatives in Representative Learning.

In summary, PBR extends Representative Learning to structured image data by combining model-independent image representation with model-aware predictive matching while retaining pseudo-images as the communication object. The simulation results show that representative-based communication can improve generalization across heterogeneous nodes without exchanging model parameters, while the ablation study highlights an important distinction between performance within the participating network and transferability of the communicated representation. Together, these results suggest that the design of distributed communication objects should balance predictive informativeness, data-space structure, and portability across models.

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Projection-Based Representative Learning for Image Classification

SUPPLEMENTARY MATERIAL

S.1 More Tables

Table S1: Class distribution across nodes

Node	Class 0	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8	Class 9	Total
0	268	0	0	234	269	107	8619	0	394	64	9955
1	0	269	0	19	11527	195	0	4374	0	0	16384
2	459	11789	12	908	0	13251	5492	1	0	14	31926
3	309	0	1	0	3	2	0	3	0	147	465
4	8413	1838	2	10179	0	65	0	430	111	2149	23187
5	2680	0	4797	55	2015	0	17	8722	0	11459	29745
6	0	3528	908	700	4	0	53	10	6955	9	12167
7	2260	0	18	3001	798	0	2	153	6340	0	12572
8	5	0	9873	0	0	274	0	3	0	0	10155
9	238	111	25	0	0	218	4	604	0	1	1201
All	14632	17535	15636	15096	14616	14112	14187	14300	13800	13843	147757

Table S2: Cross-node generalization accuracy (%). Each node row is the weighted mean accuracy of that node's model over the other nine node-specific test sets, with weights proportional to target test-set size.

Node (model)	**FedProto**	**DeSA**	**Full model**	**Ablation I**	**Ablation II**	**Ablation III**	**Ablation IV**
Shared initialization; Round 0							
0 (C)	38.81	38.81	**38.81**	38.81	38.81	38.81	**38.81**
1 (C)	28.02	28.02	**28.02**	28.02	28.02	28.02	**28.02**
2 (A)	38.40	38.40	**38.40**	38.40	38.40	38.40	**38.40**
3 (A)	17.95	17.95	**17.95**	17.95	17.95	17.95	**17.95**
4 (C)	49.35	49.35	**49.35**	49.35	49.35	49.35	**49.35**
5 (A)	49.96	49.96	**49.96**	49.96	49.96	49.96	**49.96**
6 (C)	34.78	34.78	**34.78**	34.78	34.78	34.78	**34.78**
7 (A)	32.28	32.28	**32.28**	32.28	32.28	32.28	**32.28**
8 (C)	15.28	15.28	**15.28**	15.28	15.28	15.28	**15.28**
9 (A)	24.45	24.45	**24.45**	24.45	24.45	24.45	**24.45**
Node mean	32.93	32.93	**32.93**	32.93	32.93	32.93	**32.93**
Weighted mean	30.87	30.87	**30.87**	30.87	30.87	30.87	**30.87**
100 **outer** × 10 **local; Round 1**							

Continued on the next page

Table S2: Cross-node generalization accuracy (continued)

Node (model)	FedProto	DeSA	Full model	Ablation I	Ablation II	Ablation III	Ablation IV
0 (C)	39.01	48.07	**43.80**	43.88	43.36	42.97	**47.56**
1 (C)	29.54	32.26	**38.25**	38.28	38.16	37.04	**37.21**
2 (A)	43.55	44.96	**53.88**	53.83	53.61	53.37	**52.64**
3 (A)	18.64	32.12	**16.79**	16.78	16.84	16.82	**17.24**
4 (C)	51.67	51.51	**57.79**	58.22	57.85	57.30	**57.62**
5 (A)	52.88	52.97	**60.84**	60.79	60.22	59.09	**58.61**
6 (C)	38.57	40.20	**51.87**	51.75	50.81	50.50	**51.12**
7 (A)	33.72	40.50	**34.29**	34.05	34.76	35.26	**19.17**
8 (C)	15.59	15.52	**21.59**	21.89	21.37	21.18	**26.11**
9 (A)	27.99	35.50	**19.25**	19.65	20.02	19.70	**20.44**
Node mean	35.11	39.36	**39.83**	39.91	39.70	39.32	**38.77**
Weighted mean	33.31	36.99	**38.81**	38.90	38.75	38.32	**37.64**
100 **outer** × 10 **local; Round 10**							
0 (C)	40.67	41.73	**53.81**	51.56	52.35	48.88	**62.42**
1 (C)	30.43	31.15	**64.52**	62.87	59.03	49.15	**64.01**
2 (A)	46.02	43.67	**69.98**	69.41	67.63	65.98	**68.46**
3 (A)	22.64	29.66	**18.22**	18.17	18.27	18.09	**18.33**
4 (C)	52.14	51.77	**67.68**	67.70	65.10	59.54	**75.45**
5 (A)	54.19	56.71	**69.07**	70.05	69.21	66.49	**71.08**
6 (C)	41.05	40.42	**61.78**	59.95	58.53	50.62	**66.94**
7 (A)	36.72	42.00	**48.64**	49.48	46.47	42.81	**53.45**
8 (C)	16.06	15.84	**49.26**	49.11	40.56	29.98	**57.11**
9 (A)	34.22	40.67	**36.40**	35.75	33.19	29.29	**34.93**
Node mean	37.41	39.36	**53.94**	53.40	51.03	46.08	**57.22**
Weighted mean	35.49	37.18	**52.38**	52.08	49.39	44.25	**56.19**
100 **outer** × 10 **local; Round 100**							
0 (C)	41.81	36.51	**63.36**	63.76	57.73	54.87	**80.14**
1 (C)	30.87	26.93	**59.76**	57.55	51.88	40.80	**77.28**
2 (A)	45.79	43.57	**70.72**	69.85	63.22	61.68	**82.59**
3 (A)	22.26	22.92	**33.31**	51.73	39.60	27.68	**70.70**
4 (C)	52.86	47.16	**66.80**	66.36	61.95	56.88	**84.62**
5 (A)	55.06	55.24	**71.34**	70.77	67.27	66.09	**83.00**
6 (C)	41.98	35.56	**65.17**	64.73	60.63	53.12	**79.63**
7 (A)	37.39	41.46	**60.01**	59.35	57.33	52.85	**71.96**
8 (C)	16.48	14.90	**51.73**	49.47	35.87	25.97	**79.06**
9 (A)	34.62	38.00	**56.31**	51.19	40.38	33.77	**68.06**
Node mean	37.91	36.23	**59.85**	60.47	53.59	47.37	**77.71**
Weighted mean	36.14	33.86	**59.25**	59.60	52.86	46.06	**77.91**

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Table S2: Cross-node generalization accuracy (continued)

Node (model)	FedProto	DeSA	Full model	Ablation I	Ablation II	Ablation III	Ablation IV
1000 **outer** × 1 **local; Round 1**							
0 (C)	38.11	44.90	**41.52**	41.78	41.60	41.29	**42.41**
1 (C)	27.68	34.28	**29.92**	29.91	29.87	29.91	**30.40**
2 (A)	39.39	46.63	**36.06**	36.22	35.78	35.61	**39.23**
3 (A)	18.68	18.71	**16.73**	16.72	16.85	16.78	**16.74**
4 (C)	49.77	48.21	**52.68**	52.53	52.73	52.89	**52.20**
5 (A)	50.57	52.94	**50.32**	50.02	50.98	51.96	**39.89**
6 (C)	36.04	41.65	**37.03**	37.00	36.64	36.63	**37.41**
7 (A)	31.34	38.37	**29.85**	28.96	30.24	31.75	**16.04**
8 (C)	15.12	16.32	**16.56**	16.55	16.54	16.52	**16.84**
9 (A)	23.75	27.23	**24.23**	24.26	24.26	24.38	**22.84**
Node mean	33.05	36.92	**33.49**	33.40	33.55	33.77	**31.40**
Weighted mean	31.10	34.27	**32.01**	31.95	32.13	32.37	**30.10**
1000 **outer** × 1 **local; Round 10**							
0 (C)	39.89	46.11	**49.17**	49.19	48.67	45.33	**47.02**
1 (C)	29.42	31.72	**42.47**	42.55	42.33	34.95	**38.91**
2 (A)	43.46	44.96	**56.59**	56.17	56.49	53.31	**49.40**
3 (A)	18.72	31.32	**14.78**	14.58	14.61	14.65	**17.02**
4 (C)	51.01	51.23	**60.36**	60.03	59.57	54.11	**53.72**
5 (A)	53.54	53.06	**63.90**	63.62	63.30	58.89	**55.95**
6 (C)	38.07	40.13	**43.31**	44.13	46.11	43.79	**41.23**
7 (A)	33.85	40.44	**43.15**	42.95	42.66	39.65	**27.57**
8 (C)	15.55	16.28	**28.39**	28.32	27.34	21.50	**24.75**
9 (A)	30.47	39.03	**23.36**	23.53	23.43	23.21	**22.04**
Node mean	35.40	39.43	**42.55**	42.51	42.45	38.94	**37.76**
Weighted mean	33.57	36.76	**41.18**	41.33	41.11	37.63	**36.38**
1000 **outer** × 1 **local; Round 100**							
0 (C)	40.77	43.32	**62.01**	58.85	58.38	55.25	**69.81**
1 (C)	30.41	32.44	**66.28**	65.03	60.74	46.61	**70.44**
2 (A)	45.83	43.93	**63.83**	66.61	62.07	60.55	**73.34**
3 (A)	22.47	31.05	**14.43**	14.53	14.49	14.16	**15.17**
4 (C)	52.20	51.35	**69.55**	69.40	64.81	57.96	**78.04**
5 (A)	54.23	57.28	**70.33**	71.46	71.19	68.35	**76.18**
6 (C)	40.83	39.92	**63.80**	63.79	57.75	45.84	**73.10**
7 (A)	36.89	41.28	**51.62**	50.04	49.03	48.50	**57.48**
8 (C)	16.07	16.68	**50.42**	49.99	45.25	29.42	**59.52**
9 (A)	34.11	39.14	**22.71**	22.13	20.54	21.14	**26.11**
Node mean	37.38	39.64	**53.50**	53.18	50.43	44.78	**59.92**
Weighted mean	35.50	37.44	**52.71**	52.41	49.84	43.82	**59.27**
1000 **outer** × 1 **local; Round 1000**							
0 (C)	41.76	35.03	**65.86**	65.02	60.04	53.73	**84.36**
1 (C)	30.87	27.95	**65.93**	66.54	54.88	43.35	**84.97**
2 (A)	45.57	43.69	**66.08**	67.56	56.84	52.98	**81.40**
3 (A)	22.35	23.95	**57.01**	58.38	43.86	34.38	**70.24**
4 (C)	52.80	48.76	**67.78**	68.31	63.18	56.64	**86.81**
5 (A)	54.92	54.62	**74.57**	74.83	70.16	64.53	**85.57**
6 (C)	41.81	35.84	**66.53**	68.38	59.26	51.22	**83.17**
7 (A)	37.58	42.60	**60.46**	63.56	54.82	47.62	**75.62**
8 (C)	16.48	15.17	**54.13**	56.32	38.37	26.52	**82.82**
9 (A)	34.44	39.76	**55.53**	55.31	48.30	42.41	**73.79**
Node mean	37.86	36.74	**63.39**	64.42	54.97	47.34	**80.88**
Weighted mean	36.10	34.53	**63.37**	64.20	54.48	46.39	**81.43**

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Table S2: Cross-node generalization accuracy (continued)

Node (model)	FedProto	DeSA	Full model	Ablation I	Ablation II	Ablation III	Ablation IV
10 **outer** × 100 **local; Round 1**							
0 (C)	40.58	43.26	**56.42**	55.61	55.84	54.57	**59.67**
1 (C)	30.38	29.92	**54.89**	54.82	53.27	52.16	**52.93**
2 (A)	46.09	46.15	**68.04**	68.88	67.83	67.95	**64.41**
3 (A)	22.49	26.30	**18.04**	17.95	17.95	17.92	**18.62**
4 (C)	52.28	50.51	**62.38**	62.83	61.85	60.03	**63.79**
5 (A)	54.21	57.32	**65.56**	64.63	66.32	65.38	**66.15**
6 (C)	41.02	38.98	**42.64**	42.81	43.40	41.73	**51.23**
7 (A)	36.79	44.78	**48.81**	48.04	48.77	49.19	**41.97**
8 (C)	16.10	16.04	**32.52**	32.94	31.06	29.96	**39.32**
9 (A)	34.18	41.44	**31.86**	33.44	28.80	27.47	**31.43**
Node mean	37.41	39.47	**48.11**	48.19	47.51	46.63	**48.95**
Weighted mean	35.53	37.12	**45.99**	46.08	45.69	44.80	**46.74**
10 **outer** × 100 **local; Round 10**							
0 (C)	41.79	35.55	**64.38**	63.71	59.73	57.13	**76.34**
1 (C)	30.86	26.45	**58.40**	56.89	49.72	44.20	**69.70**
2 (A)	45.86	43.65	**53.20**	53.51	52.75	48.29	**70.85**
3 (A)	22.28	21.87	**30.38**	28.66	30.44	18.74	**59.47**
4 (C)	52.84	45.31	**63.42**	64.63	61.66	58.55	**77.33**
5 (A)	55.03	55.24	**70.95**	69.64	68.40	64.93	**77.94**
6 (C)	41.79	35.26	**58.06**	58.83	55.55	50.41	**76.38**
7 (A)	37.42	42.61	**51.41**	52.65	47.40	45.68	**63.47**
8 (C)	16.48	14.86	**45.03**	44.07	34.49	29.00	**69.34**
9 (A)	34.61	37.06	**53.85**	45.25	44.03	38.13	**64.59**
Node mean	37.89	35.79	**54.91**	53.78	50.42	45.51	**70.54**
Weighted mean	36.13	33.38	**54.43**	53.29	50.03	45.94	**70.23**

Table S3: Own-node test accuracy (%). Each node row is the accuracy of that node's model on its own node-specific test set.

Node (model)	FedProto	DeSA	Full model	Ablation I	Ablation II	Ablation III	Ablation IV
Shared initialization; Round 0							
0 (C)	97.95	97.95	**97.95**	97.95	97.95	97.95	**97.95**
1 (C)	98.19	98.19	**98.19**	98.19	98.19	98.19	**98.19**
2 (A)	96.89	96.89	**96.89**	96.89	96.89	96.89	**96.89**
3 (A)	93.97	93.97	**93.97**	93.97	93.97	93.97	**93.97**
4 (C)	97.22	97.22	**97.22**	97.22	97.22	97.22	**97.22**
5 (A)	96.33	96.33	**96.33**	96.33	96.33	96.33	**96.33**
6 (C)	96.94	96.94	**96.94**	96.94	96.94	96.94	**96.94**
7 (A)	95.74	95.74	**95.74**	95.74	95.74	95.74	**95.74**
8 (C)	99.84	99.84	**99.84**	99.84	99.84	99.84	**99.84**
9 (A)	84.33	84.33	**84.33**	84.33	84.33	84.33	**84.33**
Node mean	95.74	95.74	**95.74**	95.74	95.74	95.74	**95.74**
Weighted mean	97.04	97.04	**97.04**	97.04	97.04	97.04	**97.04**
100 **outer** × 10 **local; Round 1**							

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Table S3: Own-node test accuracy (continued)

Node (model)	FedProto	DeSA	Full model	Ablation I	Ablation II	Ablation III	Ablation IV
0 (C)	98.31	98.43	**97.35**	97.31	97.15	97.31	**97.55**
1 (C)	98.66	98.34	**97.68**	97.58	97.46	97.61	**97.97**
2 (A)	97.36	97.85	**95.04**	95.19	95.18	95.08	**95.58**
3 (A)	94.83	93.10	**91.38**	91.38	91.38	91.38	**93.10**
4 (C)	97.57	97.53	**96.10**	96.24	96.26	96.10	**95.48**
5 (A)	97.31	97.12	**94.41**	94.29	94.84	95.00	**92.78**
6 (C)	97.44	97.47	**95.82**	95.92	95.86	95.96	**95.23**
7 (A)	96.40	97.49	**93.70**	93.57	93.89	94.11	**82.37**
8 (C)	99.92	99.84	**99.65**	99.68	99.68	99.68	**99.05**
9 (A)	90.67	93.33	**72.00**	73.00	73.33	72.67	**74.00**
Node mean	96.85	97.05	**93.31**	93.42	93.50	93.49	**92.31**
Weighted mean	97.63	97.77	**95.60**	95.62	95.73	95.77	**94.30**
100 **outer** × 10 **local; Round 10**							
0 (C)	98.51	98.43	**95.22**	94.49	95.30	95.74	**95.10**
1 (C)	98.66	99.00	**97.68**	97.75	97.78	97.63	**97.05**
2 (A)	97.44	98.01	**94.35**	93.74	93.77	94.73	**94.80**
3 (A)	95.69	95.69	**94.83**	94.83	95.69	94.83	**95.69**
4 (C)	97.91	97.95	**95.91**	96.17	96.12	96.17	**93.86**
5 (A)	97.55	98.16	**94.78**	94.70	94.73	94.54	**94.62**
6 (C)	97.96	97.86	**94.57**	93.95	94.77	94.38	**93.78**
7 (A)	97.45	98.35	**91.35**	89.95	91.22	90.84	**90.80**
8 (C)	99.88	99.84	**98.90**	98.90	99.21	99.61	**95.59**
9 (A)	93.67	94.33	**78.67**	80.67	79.67	77.33	**84.33**
Node mean	97.47	97.76	**93.63**	93.51	93.83	93.58	**93.56**
Weighted mean	97.92	98.27	**95.06**	94.76	95.01	95.14	**94.43**
100 **outer** × 10 **local; Round 100**							
0 (C)	98.55	97.39	**97.35**	97.51	97.43	97.11	**96.38**
1 (C)	98.73	98.22	**97.83**	97.49	97.83	98.00	**97.58**
2 (A)	97.48	98.00	**96.35**	95.16	95.83	96.24	**95.08**
3 (A)	93.10	93.97	**64.66**	76.72	57.76	94.83	**75.86**
4 (C)	97.90	96.59	**96.43**	96.46	96.67	96.69	**95.45**
5 (A)	97.59	98.10	**95.91**	95.90	96.11	95.91	**94.74**
6 (C)	97.99	97.44	**94.61**	94.25	94.57	95.76	**94.38**
7 (A)	97.65	98.19	**94.53**	93.45	93.38	94.72	**94.62**
8 (C)	99.92	99.68	**99.29**	99.57	99.65	99.65	**97.36**
9 (A)	93.00	94.33	**80.00**	76.33	81.00	77.33	**81.33**
Node mean	97.19	97.19	**91.70**	92.28	91.02	94.62	**92.28**
Weighted mean	97.95	97.82	**96.18**	95.80	96.06	96.41	**95.32**

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Table S3: Own-node test accuracy (continued)

Node (model)	FedProto	DeSA	Full model	Ablation I	Ablation II	Ablation III	Ablation IV
1000 **outer** × 1 **local; Round 1**							
0 (C)	98.27	98.07	**97.87**	97.83	97.83	97.79	**97.51**
1 (C)	98.44	98.07	**96.41**	96.39	96.29	96.22	**97.63**
2 (A)	97.06	96.56	**87.49**	87.86	87.32	86.93	**84.17**
3 (A)	94.83	95.69	**90.52**	90.52	91.38	90.52	**91.38**
4 (C)	97.41	96.33	**96.50**	96.50	95.98	95.52	**96.48**
5 (A)	96.88	96.30	**90.92**	90.45	92.01	92.87	**65.91**
6 (C)	97.04	96.74	**96.58**	96.61	96.32	96.22	**97.14**
7 (A)	96.63	95.16	**68.79**	65.83	70.44	77.86	**10.44**
8 (C)	99.80	99.80	**99.88**	99.88	99.84	99.88	**99.76**
9 (A)	85.67	85.67	**84.00**	84.00	84.00	84.33	**77.67**
Node mean	96.20	95.84	**90.90**	90.59	91.14	91.81	**81.81**
Weighted mean	97.36	96.77	**91.27**	91.00	91.48	92.11	**80.65**
1000 **outer** × 1 **local; Round 10**							
0 (C)	98.31	98.35	**91.20**	92.16	91.00	92.16	**88.83**
1 (C)	98.63	98.61	**96.97**	96.88	97.19	97.00	**97.10**
2 (A)	97.44	97.67	**94.99**	95.05	94.91	95.05	**95.78**
3 (A)	94.83	96.55	**82.76**	81.03	81.90	81.90	**93.10**
4 (C)	97.62	97.62	**94.31**	94.41	94.48	94.19	**92.72**
5 (A)	97.16	97.01	**90.65**	90.12	90.40	92.98	**88.68**
6 (C)	97.27	97.70	**94.02**	94.97	95.23	95.66	**88.06**
7 (A)	96.91	97.45	**87.81**	86.41	88.39	92.46	**87.24**
8 (C)	99.88	99.84	**99.17**	99.09	99.25	99.76	**98.42**
9 (A)	91.33	88.67	**78.67**	78.67	79.33	80.00	**73.33**
Node mean	96.94	96.95	**91.06**	90.88	91.21	92.12	**90.33**
Weighted mean	97.66	97.74	**93.40**	93.32	93.53	94.51	**92.18**
1000 **outer** × 1 **local; Round 100**							
0 (C)	98.47	98.39	**95.58**	94.73	94.65	95.34	**94.90**
1 (C)	98.61	99.07	**97.51**	97.78	98.00	98.05	**96.31**
2 (A)	97.52	98.13	**95.01**	94.78	95.23	95.14	**95.37**
3 (A)	95.69	97.41	**78.45**	78.45	78.45	75.86	**82.76**
4 (C)	97.96	97.98	**96.12**	96.14	96.22	96.29	**95.05**
5 (A)	97.59	98.01	**93.18**	92.58	93.32	93.75	**93.45**
6 (C)	97.86	97.70	**93.78**	94.05	93.49	90.56	**92.50**
7 (A)	97.58	98.15	**88.01**	89.50	89.88	89.63	**90.80**
8 (C)	99.88	99.92	**98.86**	99.05	99.17	99.68	**97.91**
9 (A)	93.67	93.67	**68.33**	65.67	62.33	64.00	**80.33**
Node mean	97.48	97.84	**90.48**	90.27	90.07	89.83	**91.94**
Weighted mean	97.95	98.26	**94.43**	94.37	94.62	94.53	**94.39**
1000 **outer** × 1 **local; Round 1000**							
0 (C)	98.55	96.99	**96.66**	96.66	96.30	95.90	**95.78**
1 (C)	98.68	98.54	**97.95**	98.19	98.27	98.34	**96.53**
2 (A)	97.49	97.92	**95.53**	95.63	96.04	96.37	**96.14**
3 (A)	93.97	95.69	**73.28**	75.86	75.86	69.83	**81.03**
4 (C)	97.90	96.86	**95.91**	96.02	96.12	96.36	**95.21**
5 (A)	97.63	98.00	**93.92**	94.37	94.66	94.35	**94.29**
6 (C)	97.80	97.34	**96.35**	95.89	96.15	96.35	**95.56**
7 (A)	97.52	98.25	**94.88**	95.07	95.13	95.13	**95.04**
8 (C)	99.92	99.72	**99.49**	99.57	99.72	99.76	**96.34**
9 (A)	92.67	93.67	**74.67**	72.33	77.33	78.33	**85.00**
Node mean	97.21	97.30	**91.86**	91.96	92.56	92.07	**93.09**
Weighted mean	97.93	97.84	**95.66**	95.78	96.01	96.05	**95.37**

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Table S3: Own-node test accuracy (continued)

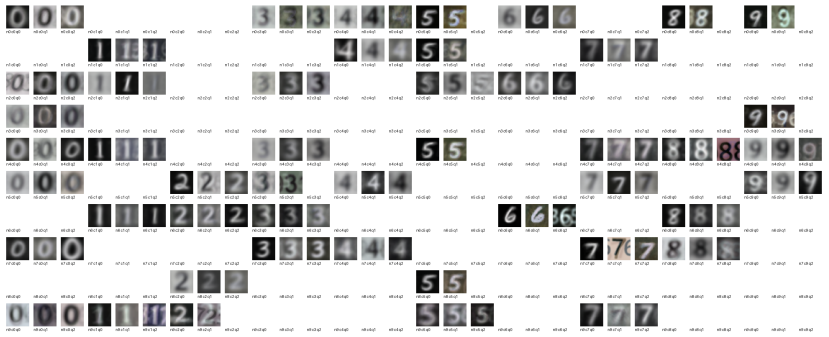
Node (model)	FedProto	DeSA	Full model	Ablation I	Ablation II	Ablation III	Ablation IV
10 **outer** × 100 **local; Round 1**							
0 (C)	98.55	98.19	**96.38**	96.26	96.10	96.02	**96.62**
1 (C)	98.68	98.95	**97.85**	97.95	98.02	97.97	**97.97**
2 (A)	97.54	98.10	**94.23**	94.28	94.31	94.10	**94.96**
3 (A)	94.83	95.69	**86.21**	85.34	85.34	85.34	**93.97**
4 (C)	97.90	97.98	**96.45**	96.40	96.27	96.21	**96.43**
5 (A)	97.61	98.01	**94.82**	94.82	94.57	94.59	**94.61**
6 (C)	97.86	98.09	**84.31**	84.91	86.81	84.41	**89.81**
7 (A)	97.45	98.03	**91.92**	92.17	92.05	92.40	**91.44**
8 (C)	99.92	99.80	**99.53**	99.53	99.53	99.49	**99.09**
9 (A)	93.67	95.33	**80.00**	79.33	81.67	84.33	**76.00**
Node mean	97.40	97.82	**92.17**	92.10	92.47	92.49	**93.09**
Weighted mean	97.95	98.24	**94.45**	94.52	94.62	94.41	**94.97**
10 **outer** × 100 **local; Round 10**							
0 (C)	98.55	97.03	**97.11**	97.11	97.11	96.91	**96.10**
1 (C)	98.71	98.00	**98.19**	98.14	98.32	98.34	**97.63**
2 (A)	97.56	98.00	**95.29**	95.67	95.64	95.48	**96.30**
3 (A)	93.10	93.10	**19.83**	49.14	21.55	8.62	**84.48**
4 (C)	97.90	96.00	**95.95**	96.02	96.12	96.95	**95.64**
5 (A)	97.61	98.06	**95.39**	95.44	95.52	95.82	**95.31**
6 (C)	97.83	97.17	**95.79**	95.56	95.79	95.99	**94.71**
7 (A)	97.58	98.28	**95.42**	94.91	94.97	95.29	**94.72**
8 (C)	99.92	99.68	**99.57**	99.61	99.68	99.76	**98.54**
9 (A)	92.67	94.33	**79.33**	68.00	72.00	68.67	**84.67**
Node mean	97.14	96.97	**87.19**	88.96	86.67	85.18	**93.81**
Weighted mean	97.95	97.66	**95.84**	95.87	95.90	96.02	**95.89**



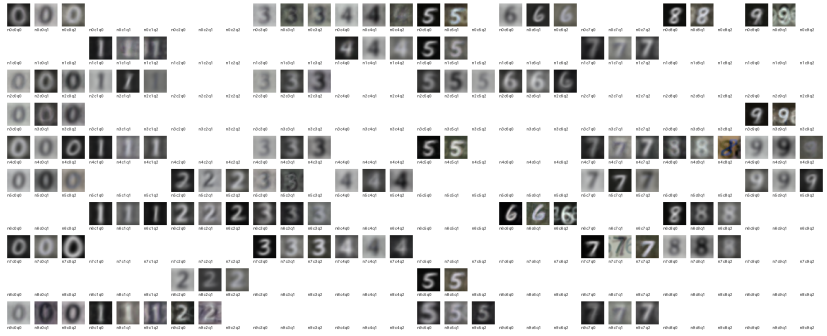
(a) Full model



(b) Ablation I



(c) Ablation II



(d) Ablation III



(e) Ablation IV

Figure S1: Representative images produced by the full model and Ablations I-IV at Round 10 under the 10-outer-round × 100-local-epoch communication protocol. Each row in a panel corresponds to one node, and the columns are organized by class and cluster. Blank cells indicate unavailable class–cluster representatives.



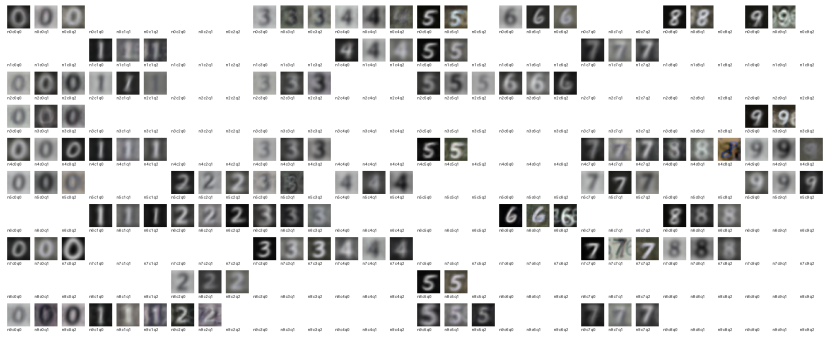
(a) Full model



(b) Ablation I



(c) Ablation II



(d) Ablation III



(e) Ablation IV

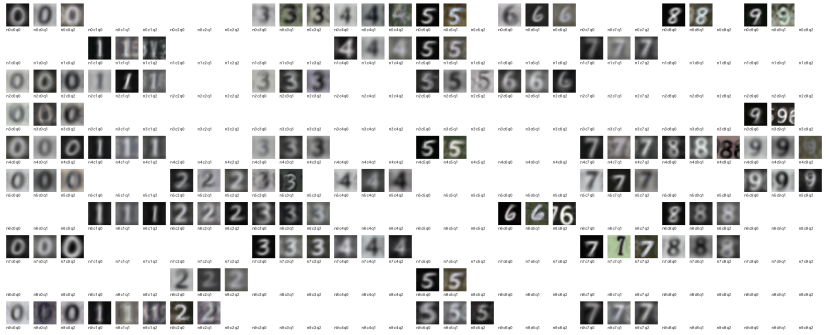
Figure S2: Representative images produced by the full model and Ablations I–IV at Round 100 under the 100-outer-round × 10-local-epoch communication protocol. Each row in a panel corresponds to one node, and the columns are organized by class and cluster. Blank cells indicate unavailable class–cluster representatives.



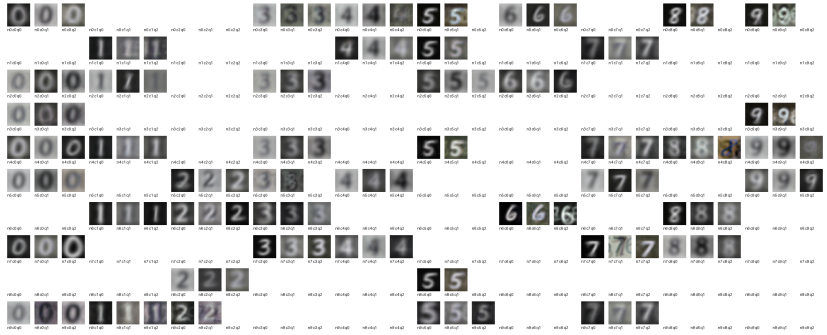
(a) Full model



(b) Ablation I



(c) Ablation II



(d) Ablation III



(e) Ablation IV

Figure S3: Representative images produced by the full model and Ablations I–IV at Round 1000 under the 1000-outer-round × 1-local-epoch communication protocol. Each row in a panel corresponds to one node, and the columns are organized by class and cluster. Blank cells indicate unavailable class–cluster representatives.